

TULLIBY LAKE AGCM, AB, Canada

WMO: 718570

Lat: 53.6644N

Lon: 110.0810W

Elev: 1955

StdP: 13.69

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 2	(b) -28.2	(c) -23.0	(d) -35.0	(e) 0.8	(f) -27.9	(g) -29.4	(h) 1.1	(i) -22.8	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	82.2	62.9	79.1	62.3	76.2	61.1	66.9	77.5	64.8	75.1	62.9	72.8	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	92.4	72.3	60.8	85.5	69.8	58.9	79.8	67.9	32.6	78.0	30.9	75.4	29.4	72.9	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) N/A	(b) N/A	(c) N/A	(e) -36.9	(f) 87.9	(g) 3.8	(h) 3.0	(i) -39.6	(j) 90.0	(k) -41.8	(l) 91.7	(m) -44.0	(n) 93.4	(o) -46.7	(p) 95.5		
(4) N/A	(5) N/A	(6) N/A	(e) -37.1	(f) 71.5	(g) 3.8	(h) 2.7	(i) -39.8	(j) 73.4	(k) -42.0	(l) 75.0	(m) -44.2	(n) 76.6	(o) -46.9	(p) 78.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.4	7.7	6.5	20.5	36.5	51.4	59.1	62.8	60.5	51.8	38.6	21.1	6.8		
	DBStd	23.60	15.52	14.20	13.81	9.62	8.07	5.61	4.67	5.47	8.03	8.61	12.36	14.77		
	HDD50	6589	1310	1217	914	414	82	7	0	1	71	367	868	1338		
	HDD65	10856	1775	1637	1379	856	424	189	98	157	400	820	1318	1803		
	CDD50	1277	0	0	0	8	126	279	398	328	125	13	0	0		
	CDD65	65	0	0	0	0	3	10	31	18	3	0	0	0		
	CDH74	1123	0	0	0	4	120	188	369	301	137	4	0	0		
	CDH80	192	0	0	0	0	17	31	62	50	32	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	17.6	0.7	0.5	0.8	0.9	1.8	3.2	3.3	2.5	1.7	0.6	0.7	0.9		
(15) Precipitation	PrecMax	22.7	1.5	1.1	1.7	2.3	5.6	5.7	8.0	6.1	3.2	1.6	1.7	2.0		
	PrecMin	11.3	0.3	0.0	0.0	0.0	0.3	1.0	0.4	0.3	0.2	0.1	0.1	0.2		
	PrecStd	3.3	0.3	0.3	0.4	0.6	1.3	1.3	1.6	1.5	0.9	0.4	0.5	0.5		
	PrecStd	3.3	0.3	0.3	0.4	0.6	1.3	1.3	1.6	1.5	0.9	0.4	0.5	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.3	39.8	52.8	72.2	82.2	84.0	85.6	85.2	84.8	71.1	51.8	40.5		
		MCWB	35.6	35.0	42.6	49.9	57.0	61.2	67.5	65.8	62.3	54.4	42.5	34.8		
	2%	DB	37.8	35.3	46.2	63.8	77.6	79.1	81.2	80.4	77.7	62.2	44.6	35.4		
		MCWB	33.8	32.0	38.2	46.8	55.1	60.4	65.9	64.6	59.0	49.8	38.4	31.6		
	5%	DB	34.6	31.9	40.7	58.2	72.4	75.4	78.0	77.2	72.6	57.2	39.3	31.2		
		MCWB	31.7	29.7	35.5	43.9	53.4	59.1	64.2	63.0	57.6	47.1	35.3	28.3		
	10%	DB	30.3	28.2	37.4	52.5	68.4	72.0	75.1	74.1	67.0	52.2	36.0	27.0		
		MCWB	27.7	26.0	33.3	40.9	51.3	57.0	62.4	61.7	54.6	44.4	33.2	24.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.8	35.4	43.6	50.3	58.9	65.2	71.0	69.6	64.0	55.3	43.6	35.4		
		MCDB	41.1	39.3	52.0	69.3	74.3	77.2	80.9	78.9	80.8	68.9	49.6	39.9		
	2%	WB	34.2	32.2	38.6	47.3	56.8	62.9	67.6	66.7	60.8	51.3	39.3	31.6		
		MCDB	37.2	34.3	44.9	62.4	71.9	74.6	78.4	77.2	75.2	60.6	44.4	35.2		
	5%	WB	31.6	29.5	35.8	44.5	54.9	61.2	65.6	64.4	58.3	47.9	35.8	28.4		
		MCDB	34.7	32.0	40.1	57.1	69.5	71.3	75.5	74.2	70.3	55.7	38.5	30.9		
	10%	WB	27.8	25.9	33.5	41.9	52.7	59.3	63.7	62.4	55.6	44.7	33.2	24.9		
		MCDB	30.4	28.3	37.2	51.2	66.2	68.6	72.9	71.9	65.8	52.0	35.8	27.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.3	19.7	18.5	20.1	24.1	21.1	21.1	22.5	23.5	18.8	15.1	15.6		
		MCDBR	19.4	21.8	21.2	30.3	30.7	27.1	26.1	28.2	33.6	27.9	19.3	20.0		
		MCWBR	16.4	19.1	15.0	16.0	13.0	11.6	13.1	14.2	16.6	16.1	13.8	16.8		
		MCDBR	19.1	21.0	20.4	29.2	27.3	22.9	24.2	25.2	31.2	25.9	17.7	19.5		
	5% WB	MCWBR	16.4	18.5	14.4	16.0	12.4	11.4	12.8	13.6	16.0	15.7	13.2	16.5		
(40) Clear-Sky Solar Irradiance	taub		0.239	0.288	0.284	0.326	0.348	0.366	0.364	0.371	0.310	0.275	0.250	0.214		
	taud		2.329	2.193	2.372	2.390	2.373	2.343	2.377	2.334	2.498	2.566	2.450	2.358		
	Ebn at Noon		242	255	288	288	285	280	278	269	277	267	239	235		
	Edh at Noon		17	28	29	33	36	37	35	35	26	19	16	14		
(44) All-Sky Solar Radiation	RadAvg		301	637	1118	1487	1835	1866	1923	1566	1102	633	297	219		
	RadStd		31	44	84	129	141	135	113	120	122	97	36	22		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page